Harsh Saxena

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SUMMARY

Innovative Research Staff Member with **3+ years of experience** in engineering **high-performance sensor integration systems**. **Architected scalable microservices** and **optimized real-time messaging pipelines**, achieving **99.9% uptime** and reducing **latency to under 50 milliseconds** for **high-security zones**. Seeking opportunities to leverage expertise in **sensor integration** and **microservices architecture** to enhance **security and communication systems** through **cutting-edge technology**.

SKILLS

Programming Languages: C++, JavaScript, Python, Java

Web Technologies: HTML, CSS, TypeScript

Frameworks and Libraries: Tailwind.CSS, NativeWind, Bootstrap, React.js, React Native, Next.js, Shadcn-UI, Node.js, Express.js.

Databases: MongoDB, MySQL, PostgreSQL, Firebase, Supabase

DevOps Tools: Docker, AWS, Jenkins, Git, GitHub, Redis, MQTT, Kafka, Microservices.

EXPERIENCE

Member of Research Staff

Bharat Electronic Limited **July 2025 - Present, Ghaziabad**

- **Engineered high-performance sensor integration pipelines** connecting **10+ critical hardware modules** (Radars, Public Address Systems, PIDS, BIDS, DIDS, step sensors, ESFP), processing **5,000+ daily telemetry events** with **99.9% uptime**.
- **Architected scalable microservices and containerized workflows using Docker**, reducing alert dispatch and event transmission latency to **under 50 milliseconds** for high-security zones like border perimeters and ammunition depots.
- **Optimized real-time messaging pipelines** utilizing **Message Queues and Redis**, boosting concurrent data handling capacity by **40%** to ensure immediate threat notification for field operators.
- **Built robust backend communication layers** for multi-vendor security hardware, enabling seamless transmission of live sensor alerts and instant automated triggering of public address systems.

App Developer

SiteGuide **February 2025 - June 2025, Remote**

- **Engineered cross-platform mobile applications** using React Native for iOS and Android, optimizing UI rendering threads and boosting overall app responsiveness by 20%.
- **Built a secure mobile notification and authentication pipeline** integrating automated OTP detection via Express.js and React Native to streamline user onboarding.
- **Implemented comprehensive localization and internationalization architectures** in React Native, dynamically managing multi-region content and location-based adaptations.
- **Integrated robust third-party payment gateways** into production mobile builds, ensuring encrypted end-to-end transactions and cutting average checkout latency by 30%.

Freelance Developer

Steploops PVT. LTD. **January 2024 - January 2025, Dehradun (Hybrid)**

- Crafted a robust CRM system to streamline internal business discussions and task management, empowering 30+ employees to efficiently oversee customer data and operations.
- **Engineered** a scalable e-commerce platform from scratch, managing **10,000+ product listings** and facilitating **real-time order processing** with **Next.js, Node.js**, and **Prisma ORM**, resulting in a **30% faster deployment** compared to traditional methods.
- **Optimized** workflows and improved operational efficiency by implementing **AWS, Jenkins**, and **Git** for continuous integration and delivery in a cross-functional team, leveraging modern DevOps practices.
- **Integrated MSG-91** for notifications and **PhonePe** for payments, delivering customized solutions with **Supabase**, **Tailwind CSS**, and agile methodologies to streamline client communication and align with business goals.

Full Stack Developer

Qviq.io **September 2023 - December 2024, Remote**

- **Boosted** team workflow efficiency in a **10+ member team**, enhancing communication and task coordination through **Gather**.
- **Automated** plan expiration data management with a **cron scheduler**, leveraging **React.js, Node.js, Tailwind CSS**, **MongoDB**, **AWS S3**, and **GPay integration** for scalable and maintainable solutions.
- Participated in daily strategy meetings to outline feature requirements and troubleshoot technical challenges, ensuring objectives aligned with project timelines using React.js and Node.js.

PROJECTS

MOM-1 Childcare & Nanny Platform

Mom Minus One Pvt. Ltd. • Active (iOS & Android)

- **Architected and deployed production-ready cross-platform mobile applications** on iOS and Android connecting parents with verified nannies across Bengaluru, featuring task-based childcare schedules and real-time location tracking.
- **Engineered secure backend workflows and automated notification systems** to manage child activity tracking and daily nanny task lists, ensuring reliable real-time coordination under strict performance constraints.

Finance-Driven CRM System

Steploops PVT LTD • February 2025 - April 2025

- **Enhanced CRM operational** efficiency by incorporating automated payment tracking, enhancing user interface design with Supabase back-end integration for increased system reliability.
- **Streamlined lead management, order tracking, and milestone-based billing** with profit margin calculations, additional item handling, internal notifications, and role-based user authentication (Admin, Store Manager, Manager).

EDUCATION

Bachelors of Technology in Computer Science & Engineering

Govind Ballabh Pant Institute of Engineering and Technology